

ISAAC ADJEI

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PROFILE

Electronic Engineering and Computer Science student at Aston University (Predicted First Class), specialising in embedded systems and hardware-software integration. Built real-world systems from circuit and firmware level through to secure full-stack applications, including embedded systems, analogue circuit designs and production-style and scalable web platforms. Seeking internships, placements and professional roles or opportunities in engineering and technology.

EDUCATION

Aston University, Birmingham, United Kingdom

Oct 2024 – Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist – Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Electronics 1, Electronics 2, Introductory Mathematics for Engineering, Power Skills

Stanmore College, London, United Kingdom

Sep 2022 – Jul 2024

Pearson BTEC Level 3 National Extended Diploma in Engineering · **D*DD (Distinction*, Distinction, Distinction)**

Best and Most Hardworking Student Award

Relevant Modules: Engineering Product Design and Manufacture, Microcontroller Systems for Engineers, Engineering Principles, Specialist Engineering Project, CAD in Engineering, Electronic Devices and Circuits, Calculus to Solve Engineering Problems, Electronic Measurement and Testing of Circuits

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, Java, PHP, SQL, MATLAB

Embedded & Hardware: Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, PCB Layout, Circuit Design, Oscilloscope, I2C / UART / SPI

Web: Next.js, React, Node.js, Tailwind CSS, PHP 8.2, MySQL, Apache

AI / ML: TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV

Cloud & DevOps: AWS, Azure, Docker, Vercel, Cloudflare, GitHub Actions

Professional: Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Time Management, Adaptability, Attention to Detail

PROJECTS

PHAEMOS: Smart Maintenance Platform

2026, ongoing

FastAPI, Python, Next.js 14, PostgreSQL, Redis, scikit-learn, Docker, ESP32, Arduino, STM32

[GitHub ↗](#) | [Website ↗](#)

- End-to-end IoT platform: ESP32, Arduino and STM32 nodes POST telemetry to FastAPI (PostgreSQL, Redis) in **under 200ms**; Next.js dashboard polls every 5s with live Recharts charts and real-time anomaly highlighting
- Unsupervised Isolation Forest scores every reading; above 0.7 triggers alert and auto-generates a maintenance ticket, above 0.85 attaches a diagnostic recommendation
- **JWT auth with role-based access control** (Admin, Technician, Viewer) at API and UI level; full **immutable audit log**; targeting industry-scale deployment

Two-Stage Audio Amplifier

2026

Analogue Design, PCB Design, KiCad, Proteus, Electronics

[Portfolio ↗](#)

- Designed and built a two-stage amplifier using a TL071 active band-pass filter and OPA551 output buffer
- Delivered a **full engineering workflow: simulation, breadboard validation, PCB layout and bench testing**
- Validated performance with measured output of **2.980 Vpp** against a **3 Vpp** design target

4x4x4 NeoPixel LED Cube

2025

Arduino, C++, WS2812B, Embedded Systems

[GitHub ↗](#) | [Portfolio ↗](#)

- **Non-blocking state machine architecture:** four animation modes, LDR adaptive brightness and debounced controls with serial diagnostics and electrical protection

AstonCV – Full-Stack CV Database

2026

PHP 8.2, MySQL, CSS, Apache, Cloudflare

[GitHub ↗](#) | [Portfolio ↗](#)

- Built from scratch with no frameworks; **11 security controls** including **bcrypt**, **CSRF tokens**, **PDO prepared statements** and **brute-force protection**
- **Deployed live** at Aston University with searchable profiles, filtering and server-side PDF export

CNC Milling Machine Control System

2024

Arduino, C++, Safety Systems, State Machines

[Portfolio ↗](#)

- Implemented a **safety-focused controller** using **finite state machine logic** and strict transition validation
- Integrated door interlocks, **emergency stop handling**, timed cooldown and LCD status feedback for safe operation

git-unlocked – Open Source Git Course

2026

Git, GitHub, GitLab, Markdown, Open Source

[GitHub ↗](#) | [Portfolio ↗](#)

- **170+ topic files** covering Git and all major platforms (GitHub, GitLab, Bitbucket, Azure DevOps) with cross-platform guidance from fundamentals to advanced workflows
- Maintained open-source contributor workflow: issue templates, PR standards and CI

EXPERIENCE

Student Representative

Aston Students' Union, Birmingham, UK

Student advocacy, stakeholder communication, academic liaison

Sep 2025 – Present

- Collect and communicate student feedback to academic staff through SSLC, Senate and Council meetings
- Delivered improvements to learning delivery and the wider course experience

Engineering Operations & Maintenance Planning (Virtual Experience)		British Airways via Forage, Remote
<i>Maintenance planning, risk analysis, reporting</i>		<i>Oct 2025</i>
- Analysed A320 maintenance schedules, diagnosed component faults and produced structured WREQ engineering reports - Developed material forecasts and maintenance planning reports for six aircraft with risk-mitigation recommendations		
Smart Mobility & Digital Tech (Virtual Work Experience)		Yunex Traffic via Springpod, Remote
<i>Intelligent transport systems, sensor integration, data-driven urban mobility</i>		<i>Aug 2025</i>
- Analysed intelligent transport systems and digital solutions to evaluate their impact on air quality and urban mobility efficiency - Delivered technical insight outputs based on real-time pollution sensing and smart traffic workflows		
Consular & Administrative Intern		Ghana High Commission, London, UK
<i>Consular operations, document processing, facilities coordination</i>		<i>Mar – Aug 2024</i>
Managed passport/visa processing and document verification; coordinated facilities and administrative operations		
Junior Apprentice HVAC Technician		Massive Refrigeration Services, Accra, Ghana
<i>Field servicing, diagnostics, installation support</i>		<i>Jul 2019 – Jul 2021</i>
Serviced, installed and maintained 50+ air conditioning units using standard diagnostic tools		
VOLUNTEERING		
Student Judge – National Emerging Talent Awards 2026		<i>Feb 2026</i>
<i>targetjobs UK (Group GTI)</i>		
- Evaluated employer submissions for Best Placement or Internship Programme using qualitative and scoring criteria - Commended for a thorough, timely and high-quality approach		
Fundraising Volunteer – 10 Days of 5K Challenge		<i>Feb – Mar 2026</i>
<i>Cancer Research UK</i>		
Completed 10 x 5km (50km+) for Cancer Research UK; managed fundraising page contributing to a campaign that raised over £797,000		
INTERESTS & HOBBIES		
Outside Engineering (Piano, Music, Gym & Fitness, Cycling, Cooking, Reading, Journaling, Travel), Embedded Systems, IoT, Hardware-Software Integration, Cloud Computing, Artificial Intelligence & ML, Cyber Security, Robotics & Automation, Circuit Design & PCB, Open Source, Accessible Technology, Computer Systems, Game Development		
SPOKEN LANGUAGES		
English (Full professional), Twi & Ga (Full Native), French & Spanish (Elementary)		